

Military Asset Management System (MAMS)

Comprehensive System Manual • End-to-End Workflow • Role Security Clearance

1. Executive Overview & System Purpose

The **Military Asset Management System (MAMS)** is a secure enterprise defense application designed for military commanders and logistics officers to track, manage, and audit critical assets—including **heavy vehicles, weapons, ammunition, and tactical protective gear**—across distributed bases with strict accountability and Zero-Trust Role-Based Access Control (RBAC).

Core Problems Solved:

- **Inventory Drift & Phantom Stock:** Traditional spreadsheets lead to inaccurate counts during deployments.
- **Accountability & Chain of Custody:** Assets must be traceable from procurement, through inter-base transfers, down to the exact soldier or officer in custody.
- **Operational Readiness:** Commanders need instant derived visibility into opening balances, daily net movements, and expended combat materials.
- **Tamper-Proof Audit Integrity:** Financial ledger immutability prevents unauthorized record alteration or deletion.

2. Technology Stack & Key Architectural Decisions

Component	Technology	Architectural Responsibility
Backend API	Java 17 / Spring Boot 3.3.4	Stateless REST APIs, Spring Security 6, transactional business logic, Flyway DB migrations.
Database	MySQL 8.0 (InnoDB)	Relational integrity with foreign keys, checks, unique constraints, and ACID transactions.
Security & Auth	JWT (Stateless Bearer)	384-bit HMAC-SHA signing, UserPrincipal context, method-level @PreAuthorize.
Audit Trail	Spring AOP (AspectJ)	Automatic interception of all mutating actions, logging IP, user, endpoint, and status.
Frontend UI	React 19 / Vite / Tailwind	Modern Tactical Command dashboard, role-aware routing, modal dialogs, and filters.

3. The Core Balance Calculation Engine (Critical Rule)

A foundational principle of MAMS is that **asset balances are DERIVED in real-time** and *never stored as a mutable counter*. Storing mutable counters introduces balance drift, race conditions, and ledger corruption. Instead, balances are calculated on demand from immutable ledger transactions:

Formula Component	Mathematical Definition
Net Movement	net_movement = purchases + transfers_in - transfers_out
Opening Balance	opening_balance = closing_balance as of (start_date - 1 day) Cumulative sum of all past purchases + transfers_in - transfers_out - expended prior to the date range.
Closing Balance	closing_balance = opening_balance + net_movement - expended
Assigned Count	Active custody count: items currently assigned to personnel that have not yet been expended or returned.

4. Step-by-Step Feature Walkthrough

Step 1 — Authentication & Session Management

Users authenticate via **POST /api/auth/login** using username and password. Upon successful validation, the backend issues a signed JWT containing *userId*, *role*, *baseId*, and *baseName*. The frontend stores this token and includes it in all subsequent requests via Axios HTTP interceptors. Zero public registration exists—all accounts are strictly provisioned by an Administrator.

Step 2 — Command Dashboard & Net Movement Popup

The dashboard presents high-level operational intelligence for any selected date range and base. Commanders can click the **Net Movement** KPI card to open an interactive modal breakdown showing every inbound purchase shipment and transfer with timestamps, equipment names, and source/destination bases.

Step 3 — Purchases & Procurement Ledger

Authorized personnel record new asset deliveries into a base. Each purchase records quantity, unit cost, total cost, vendor, purchase date, and notes. **Ledger Immutability:** PUT and DELETE endpoints are intentionally omitted to preserve financial ledger and legal audit integrity.

Step 4 — Inter-Base Asset Transfers

Enables moving assets between military installations with a strict 4-state lifecycle machine: **pending** → **in_transit** → **completed** / **cancelled**. Non-admins can only initiate or manage transfers where their assigned installation is either the sender or receiver.

Step 5 — Personnel Assignments & Expenditures

Tracks physical custody when equipment is checked out to soldiers and officers. Assets transition through 3 statuses: **assigned** (in personal custody) → **expended** (consumed in firing drills/combat) or **returned** (returned to armory after inspection). **Strict Policy:** Logistics Officers are completely barred from this module at both API and UI levels.

Step 6 — Personnel Identity & Access Control (User Management)

An Administrator-only console to create accounts, update clearance roles, reset passwords without old credentials, and soft-deactivate operators. A critical safety guard prohibits administrators from deactivating or demoting their own account.

5. Role-Based Access Control (RBAC) Clearance Matrix

MAMS enforces strict least-privilege security across three distinct military personnel tiers:

Capability / Permission	ADMIN	BASE_COMMANDER	LOGISTICS_OFFICER
Base Visibility Scope	Global (All Bases)	Assigned Base Only	Assigned Base Only
View Dashboard Metrics	Full Access (Any Base)	Own Base Metrics Only	Own Base Metrics Only
Record New Purchases	For Any Base	For Own Base Only	For Own Base Only
View Purchases History	Enterprise-wide	Own Base Only	Own Base Only
Initiate Inter-Base Transfers	Between Any Two Bases	Must Involve Own Base	Must Involve Own Base

Update Transfer Status	Allowed	Allowed (Own Base)	Allowed (Own Base)
Assign Assets to Personnel	Allowed (All Bases)	Allowed (Own Base)	DENIED (HTTP 403)
Mark Expended / Returned	Allowed (All Bases)	Allowed (Own Base)	DENIED (HTTP 403)
User Management Console	Full Admin Console	DENIED (HTTP 403)	DENIED (HTTP 403)

6. Ready-to-Use Test Accounts & Verification Guide

To verify all features and role restrictions immediately, log in to **http://localhost:3000** using these seeded accounts:

Role	Username	Password	Assigned Installation	Expected UI Behavior
ADMIN	admin	Admin@123	Global (None)	All 5 menu tabs visible. Global base dropdowns. Access to User Management.
BASE_COMMANDER	johndoe	Password@123	Fort Alpha	Can assign and expend assets. Scoped to Fort Alpha. User Management hidden.
LOGISTICS_OFFICER	jmiller	Password@123	Fort Alpha	Assignments & Users tabs HIDDEN and blocked with 403 on direct URL.

7. Guided 5-Minute Testing Walkthrough

- **Step 1: Inspect the Command Dashboard:** Log in as **admin**. Notice the live derived balances: Opening Balance (25,365), Net Movement (+10,107), Closing Balance (31,772). Click the **Net Movement** card to see the inbound/outbound breakdown.
- **Step 2: Record an Asset Purchase:** Navigate to **Purchases**. Click **'Record Purchase'**. Buy 20 M4 Carbines for Fort Alpha from Colt Defense. Watch the table update immediately.
- **Step 3: Dispatch an Inter-Base Transfer:** Navigate to **Transfers**. Transfer 5 Humvees from Fort Alpha to Fort Bravo. Transition the status from *pending* to *in_transit*, then *completed*.
- **Step 4: Arm Personnel & Fire Ammunition:** Log in as **johndoe** (Base Commander). Go to **Assignments & Expenditures**. Assign 500 rounds of ammo to a Sergeant. Click **'Expend'** to log live-fire usage. Check the Dashboard: the Expended counter increments automatically.
- **Step 5: Verify Logistics Officer Defense Guard:** Log in as **jmiller** (Logistics Officer). Notice that **Assignments** and **User Management** are completely missing from the sidebar. Manually navigating to */assignments* redirects to */dashboard*.
- **Step 6: User Administration & Lockout Safety:** Log back in as **admin**. Open **User Management**. Try deactivating your own account: the Deactivate button is disabled with a safety tooltip preventing administrative lockout.

8. Database Entity Relationship Overview

Entity Table	Key Columns & Foreign Keys	Purpose
bases	id, name, location, is_active	Physical military installations (Fort Alpha, Fort Bravo, Fort Charlie)
equipment_types	id, name, category, unit	Standard catalog (vehicle, weapon, ammunition, other)
users	id, username, email, password_hash, role, base_id, is_active	Operators, credentials, and base associations
purchases	id, base_id, equipment_type_id, quantity, unit_cost, total_cost, vendor, purchase_date, created_by	Immutable ledger of inbound procurements
transfers	id, from_base_id, to_base_id, equipment_type_id, quantity, status, transfer_date, created_by	Inter-base logistics tracking with state machine
assignment_expenditures	id, base_id, equipment_type_id, personnel_name, personnel_id_no, quantity, status, assigned_date, expended_date, returned_date, created_by	Armory checkouts, personal custody, and ammunition consumption
audit_logs	id, user_id, action, entity_type, entity_id, method, endpoint, status_code, details_json, ip_address, timestamp	AOP automated compliance and security log